LESSON PLAN

PISTON ENGINE (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Heat treatment of Iron and its alloys
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Heat treatment of Iron and its alloys. Advantages, procedure and different types of heat treatment
5. Specific Objectives	18.1.3.1 Trainee will be able to understand and know the heat treatments of Iron and alloys.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of heating and quenching procedures with safety precautions; Instructor supervision required
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is heat treatment?
(b) Why is heat treatment necessary for iron alloys?
(c) List the basic steps of a heat treatment cycle.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and purpose of heat treatment • Advantages of heat treatment processes • General procedure and equipment used	L & Ds	PPT, IB	15 min
(b)	• Annealing process and applications • Normalizing process and grain structure • Hardening and quenching techniques	L & Ds	PPT, PDF	20 min
(c)	• Tempering process and stress relief • Comparison of different heat treatment types • Selection criteria for specific alloys	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between annealing and normalizing.
(b) Describe the role of quenching in the hardening process.

12. Home Assignment and Follow Up	Prepare a brief note on how heat treatment affects the mechanical properties of piston engine components, linking to the stress concepts covered in LP 09.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs

1	Introduction to Stress & Strain, various types of stress and strain, explanation using stress strain diagram of various metals, Hooks law, factor of safety and other terminologies.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min